

SIMATS ENGINEERING

ASSESSMENT TOOL I — Uncertainty Visualization Lab

Course: Data Handling and Visualization	Code: DSA0601	CO: C05
Duration: 180 min	Total Marks: 100	Reg No : 192524118 Name: R. Akshaya

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Uncertainty Visualization Lab

Weightage: 20%

Code of Conduct:

I R. AKSHAYA (192524118)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. AKSHAYA

Criteria	Understanding of Uncertainty 25%	Visualization of Application of Uncertainty 25%	Organization of Visualization Principles 20%	Accuracy 10%	Clarity 5%	Total
Q1						
Q2						
Q3						

Q4

Q5

Uncertainty Visualization Lab – Assessment Tool

R Programming Practical Questions with Datasets

Instructions: Answer all five questions. Use R programming to construct the required visualization, interpret uncertainty, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

Q1. Bootstrapping and Confidence Intervals – Point Estimate

Question

A quality-control team records the following delivery times (in minutes) for 12 randomly selected orders. Using R, calculate the sample mean and obtain a 95% bootstrap confidence interval for the mean using at least 5,000 bootstrap resamples. Visualize the bootstrap distribution and mark the observed mean.

Data:

31, 35, 28, 42, 37, 33, 45, 29, 39, 34, 41, 36

Expected R tasks: Resampling, point estimate, confidence interval, histogram/density plot, interpretation.

Aim

To estimate the mean delivery time and quantify its uncertainty using bootstrap resampling, a 95% bootstrap confidence interval, and a visualization of the bootstrap distribution.

R Code

```
library(ggplot2)

delivery <- c(31,35,28,42,37,33,45,29,39,34,41,36)

set.seed(123)

observed_mean <- mean(delivery)

bootstrap_means <- replicate(5000,mean(sample(delivery,replace=TRUE)))

ci <- quantile(bootstrap_means,c(0.025,0.975))

print(observed_mean)

print(ci)
```

```
bootstrap_data <- data.frame(mean=bootstrap_means)

ggplot(bootstrap_data,aes(x=mean))+

geom_histogram(aes(y=after_stat(density)),bins=30,fill="skyblue",color="black")+

geom_density(linewidth=1)+

geom_vline(xintercept=observed_mean,color="red",linewidth=1.2)+

labs(title="Bootstrap Distribution of Mean Delivery Time",
x="Bootstrap Sample Mean",
y="Density")+

theme_minimal()
```

Interpretation

The sample mean represents the observed average delivery time of the 12 randomly selected orders.

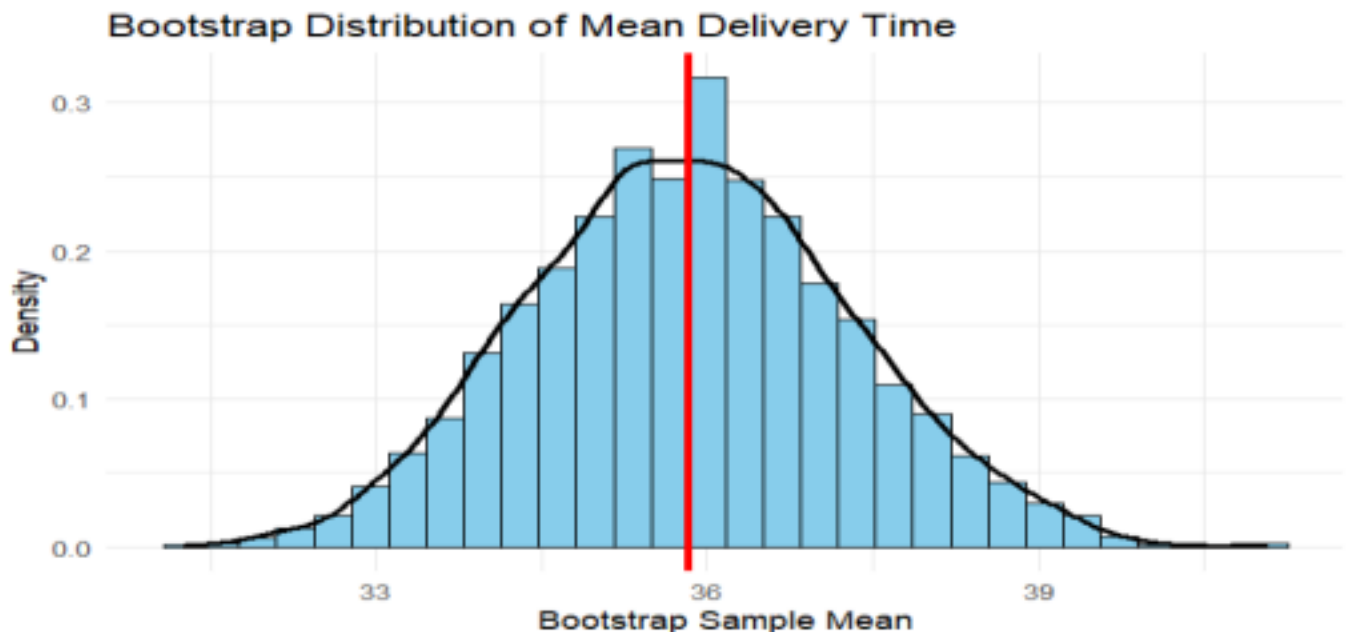
The bootstrap procedure repeatedly samples the original observations with replacement and calculates the mean for each resample. The resulting distribution shows how the sample mean can vary from sample to sample.

The 95% bootstrap confidence interval provides a range of plausible values for the population mean delivery time. Approximately 95% of the bootstrap estimates are expected to fall within this interval.

Uncertainty Explanation

The spread of the bootstrap distribution represents sampling uncertainty in the estimated mean. A narrow distribution indicates that the mean is estimated relatively precisely, while a wider distribution indicates greater uncertainty.

Visualization Explanation



The histogram displays the distribution of bootstrap means, the density curve shows its overall shape, and the red vertical line identifies the observed sample mean.

Q2. Uncertainty in Trend Estimation

Question

The following data represent monthly advertising expenditure and sales. Fit a linear regression model in R to estimate the sales trend. Plot the observations with the fitted regression line and a 95% confidence band. Explain how the uncertainty changes across the range of advertising expenditure.

Data:

Advertising:

10, 12, 15, 18, 20, 23, 25, 28, 30, 33, 36, 40

Sales:

48, 51, 53, 57, 60, 61, 66, 69, 72, 75, 79, 84

Expected R tasks: `lm()`, prediction/confidence interval, `ggplot2` visualization, trend interpretation.

Aim

To estimate the relationship between advertising expenditure and sales using linear regression and visualize the uncertainty around the estimated sales trend using a 95% confidence band.

R Code

```
library(ggplot2)
```

```

data <- data.frame(
  Advertising=c(10,12,15,18,20,23,25,28,30,33,36,40),
  Sales=c(48,51,53,57,60,61,66,69,72,75,79,84)
)

model <- lm(Sales~Advertising,data=data)

print(summary(model))

pred <- predict(model,interval="confidence")

data$Fit <- pred[,1]

data$Lower <- pred[,2]

data$Upper <- pred[,3]

ggplot(data,aes(x=Advertising,y=Sales))+
  geom_point(size=3)+
  geom_line(aes(y=Fit),linewidth=1)+
  geom_ribbon(aes(ymin=Lower,ymax=Upper),alpha=0.25)+
  labs(title="Advertising Expenditure and Sales",
        x="Advertising Expenditure",
        y="Sales")+
  theme_minimal()

```

Interpretation

The scatterplot and fitted regression line show a positive relationship between advertising expenditure and sales.

As advertising expenditure increases, sales generally increase. Therefore, the fitted regression line represents an upward sales trend.

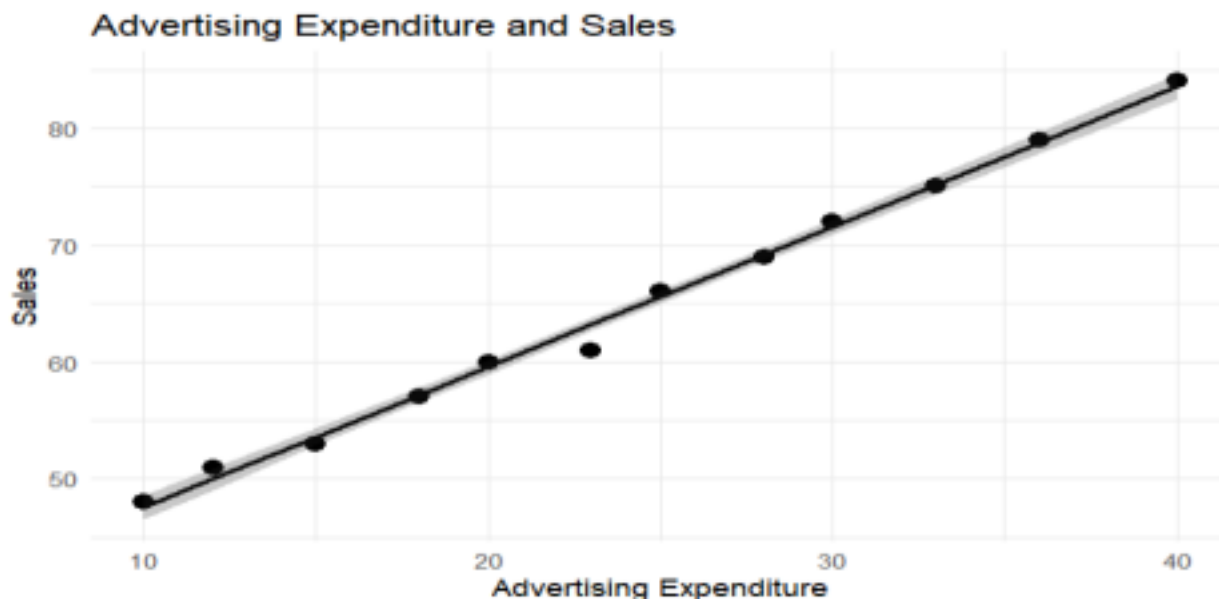
The shaded region represents the 95% confidence band around the estimated mean sales.

How Uncertainty Changes

The confidence band is generally narrower near the center of the observed advertising range and becomes wider toward the extreme values.

This occurs because there is more information available near the center of the predictor range, while estimates near the boundaries have greater uncertainty.

Visualization Explanation



The points represent the actual observations, the line represents the estimated regression trend, and the shaded region represents uncertainty in the estimated mean response.

Q3. Overplotting, Jittering and Transparency

Question

A survey records the number of hours studied and examination scores for 30 students. Several students have identical values. Create a scatterplot in R first without modification, then redesign it using partial transparency and jittering. Compare the two plots and explain how overplotting affects interpretation. **Data:**

Hours:

2,3,3,4,4,4,5,5,5,5,6,6,6,6,6,7,7,7,7,7,8,8,8,8,9,9,9,10,10

Scores:

45,52,52,58,60,60,64,66,66,66,66,70,72,72,72,72,76,78,78,80,80,84,84,85,85,85,88,90,90,94,94

Expected R tasks: `geom_point()`, `alpha`, `position_jitter()`, comparison of overplotting. **Aim**

To identify the effect of overplotting in a scatterplot and improve data visibility using transparency and jittering.

R Code

```
library(ggplot2)
```

```
data <- data.frame(
```

```

Hours=c(2,3,3,4,4,4,5,5,5,5,6,6,6,6,7,7,7,7,8,8,8,8,9,9,10,10),
Scores=c(45,52,52,58,60,60,64,66,66,66,70,72,72,72,72,76,78,78,80,80,84,84,85,85,85,88,90,90,94,94))

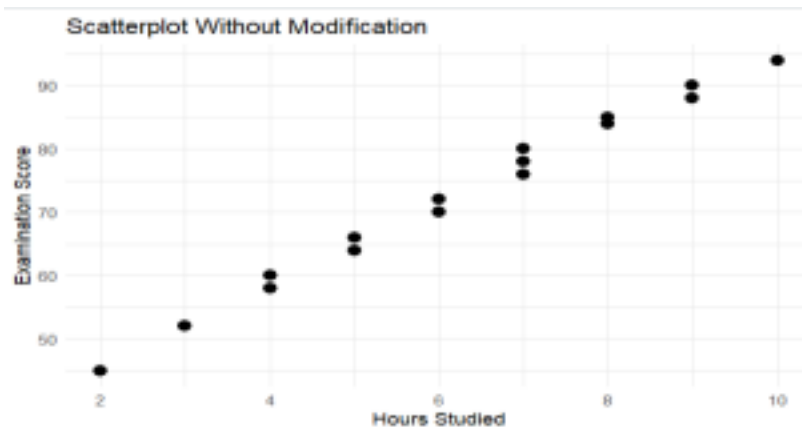
ggplot(data,aes(x=Hours,y=Scores))+
  geom_point(size=3)+
  labs(title="Scatterplot Without Modification",
    x="Hours Studied",
    y="Examination Score")+
  theme_minimal()

ggplot(data,aes(x=Hours,y=Scores))+
  geom_point(alpha=0.5,size=3)+
  labs(title="Scatterplot Using Transparency",
    x="Hours Studied",
    y="Examination Score")+
  theme_minimal()

ggplot(data,aes(x=Hours,y=Scores))+
  geom_jitter(width=0.15,height=0.5,alpha=0.6,size=3)+
  labs(title="Scatterplot Using Jittering",
    x="Hours Studied",
    y="Examination Score")+
  theme_minimal()

```

Interpretation

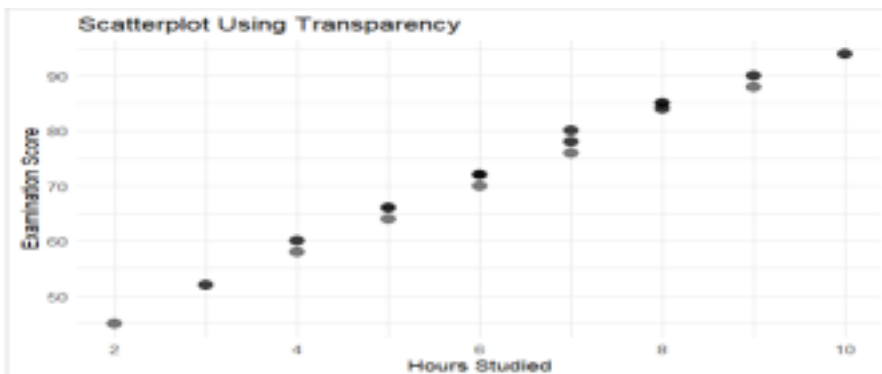


In the first scatterplot, several observations overlap because multiple students have the same number of study hours and similar examination scores. This is called overplotting.

Overplotting can make it difficult to determine:

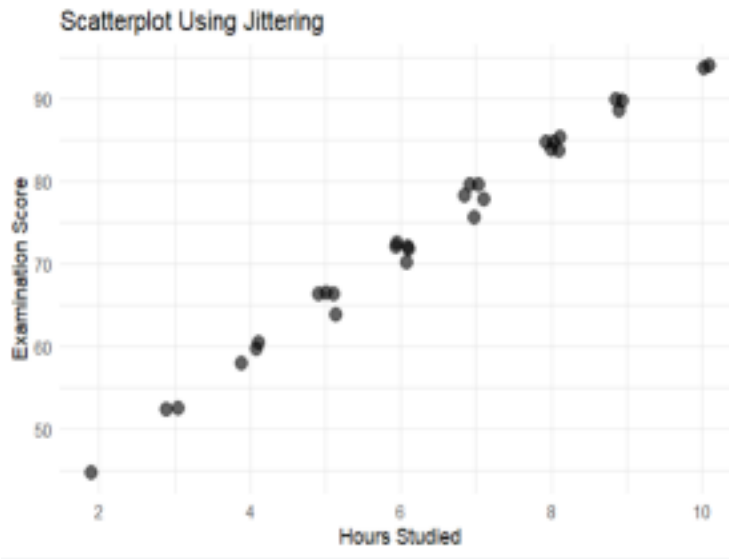
- How many observations exist at a particular location.
- Where observations are concentrated.
- The actual distribution of the data.

Transparency



Using $\alpha=0.5$ makes overlapping points partially transparent. Areas with many overlapping observations appear darker, making concentration easier to identify.

Jittering



`position_jitter()` introduces a small random displacement to the points. This separates observations that would otherwise be drawn directly on top of one another.

Comparison

Plot	Advantage	Limitation
Original scatterplot	Shows actual values	Overlapping points are difficult to distinguish
Transparency	Shows concentration through darker regions	Heavy overlap may still be difficult
Jittering	Separates overlapping observations	Points are slightly displaced from exact positions

Conclusion

The redesigned plots provide a clearer representation of the observations. Transparency and jittering reduce the visual problems caused by overplotting without changing the underlying data values conceptually.

Q4. 2D Histogram and Density Estimation Question

A researcher measures height and weight for 20 participants. Construct a 2D histogram and a 2D density visualization in R. Use the plots to identify regions with high observation density. State which visualization communicates concentration more clearly and justify your choice.

Data:

Height:

150,152,155,158,160,161,163,165,166,168,170,171,173,175,176,178,180,182,184,186

Weight:

48,50,53,55,57,59,60,62,64,65,68,69,71,73,74,77,80,82,85,88

Expected R tasks: 2D binning, density estimation, contour/density layers, interpretation of concentration.

Aim

To visualize the joint distribution of height and weight using a 2D histogram and 2D density estimation and identify regions where observations are concentrated.

R Code

```
library(ggplot2)

data <- data.frame(

  Height=c(150,152,155,158,160,161,163,165,166,168,170,171,173,175,176,178,180,182,184,186),

  Weight=c(48,50,53,55,57,59,60,62,64,65,68,69,71,73,74,77,80,82,85,88)

)

ggplot(data,aes(x=Height,y=Weight))+

  geom_bin2d(bins=6)+

  labs(title="2D Histogram of Height and Weight",

  x="Height",

  y="Weight")+

  theme_minimal()

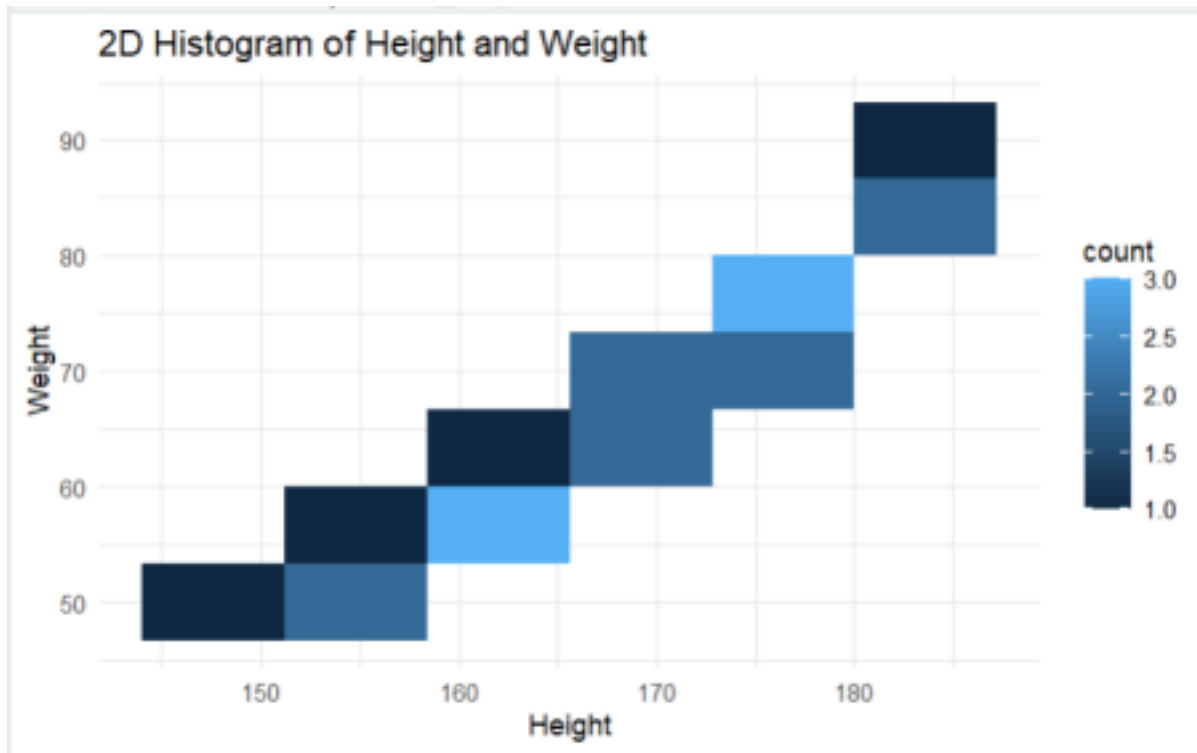
ggplot(data,aes(x=Height,y=Weight))+

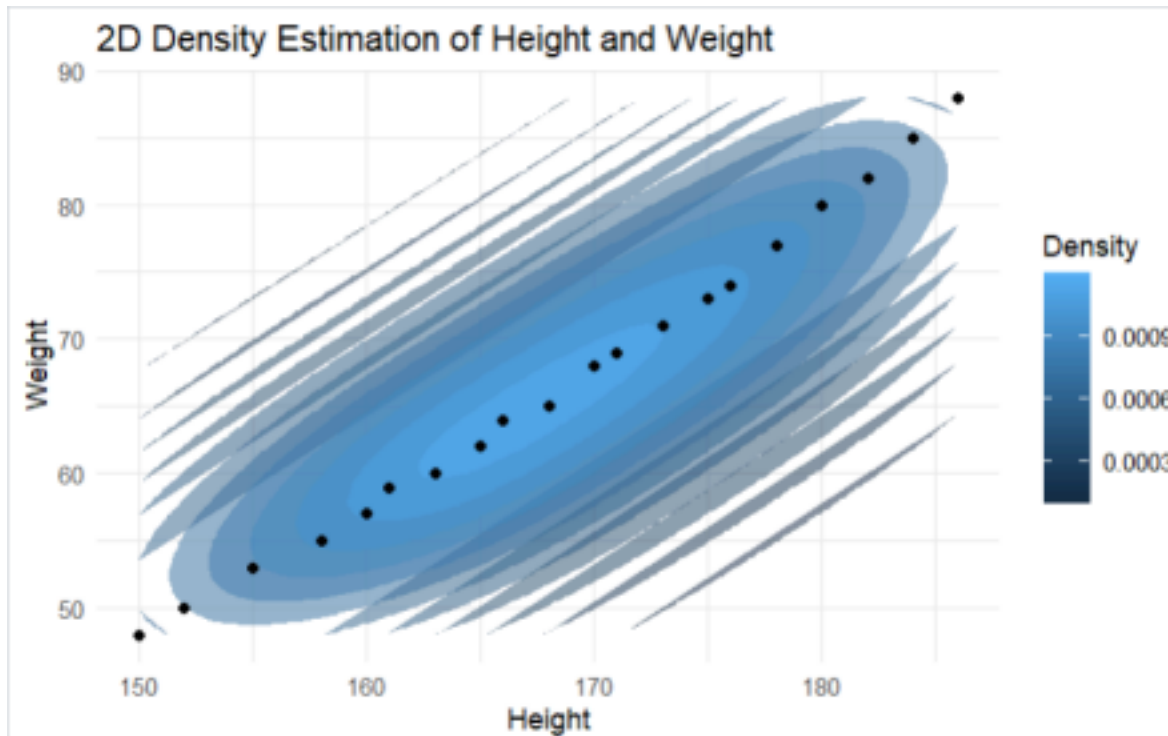
  stat_density_2d(aes(fill=after_stat(level)),geom="polygon",alpha=0.5)+

  geom_point()+
```

```
labs(title="2D Density Estimation of Height and Weight",  
x="Height",  
  
y="Weight",  
  
fill="Density")+  
  
theme_minimal()
```

Interpretation





The 2D histogram divides the height-weight space into rectangular bins. Each bin represents the number of observations within that region.

Regions containing more observations represent areas of higher concentration.

The 2D density plot estimates a smooth probability density over the height-weight space. Higher-density regions indicate combinations of height and weight where observations are more concentrated.

Which Visualization Is Better?

The 2D density visualization communicates concentration more clearly because it provides a smoother representation of the distribution and makes the high-density regions easier to identify.

The 2D histogram is useful for directly showing counts in discrete rectangular regions, whereas the density plot gives a continuous representation of concentration.

Conclusion

Both visualizations are appropriate, but the density plot is preferable when the primary objective is to understand the overall concentration and shape of the joint distribution.

Q5. Color Scales, Accessibility and Misleading Visualization

Question

The following category-wise satisfaction scores must be visualized in R. Create two plots: 1. A plot using a nonmonotonic/diverging color scheme that could mislead interpretation. 2. A redesigned plot using an ordered, color-vision-deficiency-friendly sequential scale.

Explain why the second design is more effective and identify the visualization pitfall in the first design.

Data:

Category: A, B, C, D, E, F, G

Satisfaction: 42, 48, 55, 61, 67, 74, 82

Expected R tasks: Color mapping, scale selection, accessibility-aware redesign, critique of misleading color use.

Aim

To demonstrate how inappropriate color scales can mislead interpretation and redesign the visualization using an ordered, color-vision-deficiency-friendly sequential color scale.

R Code

```
library(ggplot2)

library(viridis)

data <- data.frame(

  Category=c("A","B","C","D","E","F","G"),

  Satisfaction=c(42,48,55,61,67,74,82)

)

ggplot(data,aes(x=Category,y=Satisfaction,fill=Satisfaction))+

  geom_col()+

  scale_fill_gradient2(

    low="blue",

    mid="white",

    high="red",

    midpoint=61

  )+

  labs(

    title="Potentially Misleading Diverging Color Scale",

    x="Category",
```

```

y="Satisfaction",

fill="Satisfaction"

)+

theme_minimal()

ggplot(data,aes(x=Category,y=Satisfaction,fill=Satisfaction))+

geom_col()+

scale_fill_viridis_c(option="viridis")+

labs(

title="Accessibility-Friendly Sequential Color

Scale", x="Category",

y="Satisfaction",

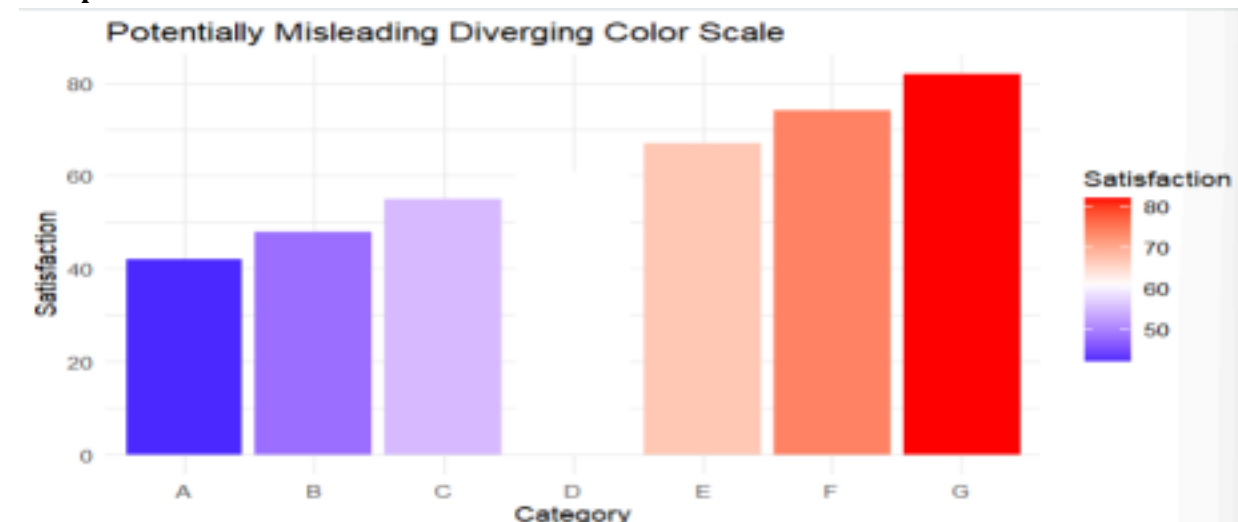
fill="Satisfaction"

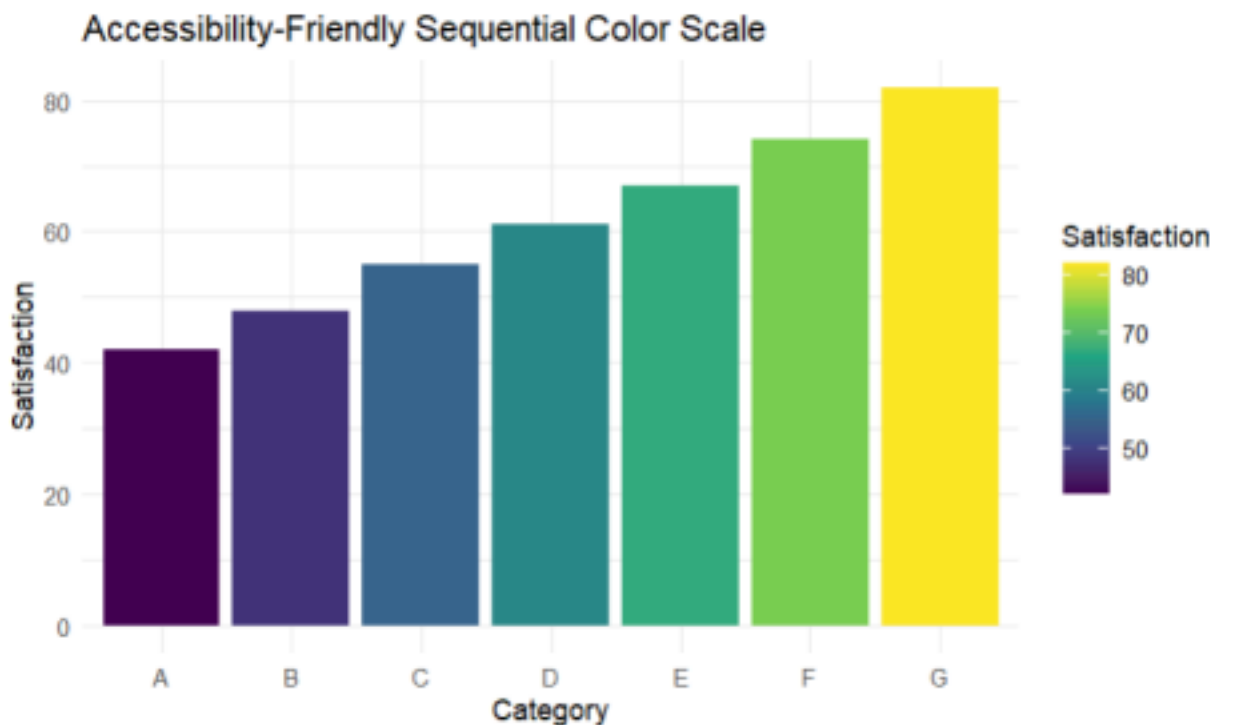
)+

theme_minimal()

```

Interpretation





In the first plot, a diverging color scale is used. It introduces different visual meanings such as blue, white, and red around an artificial midpoint.

This can be misleading because the data represent a single ordered quantitative variable. There is no natural meaningful central threshold in the data that requires two contrasting color directions.

The color changes may cause viewers to perceive categories around the midpoint as fundamentally different even though satisfaction is simply increasing.

Why the Second Plot Is Better

The second plot uses the Viridis sequential color scale.

A sequential scale is appropriate because satisfaction has a natural low-to-high ordering. The Viridis scale also has strong accessibility characteristics and is designed to remain distinguishable for many viewers with color-vision deficiencies.

Visualization Pitfall in the First Plot

The main pitfall is inappropriate color encoding.

A diverging scale should generally be used when values have a meaningful midpoint or reference value, such as:

- Below/above target
- Negative/positive
- Decrease/increase

Here, the satisfaction scores simply progress from low to high. Therefore, a sequential scale communicates the ordering more accurately.

Conclusion

The second visualization is more effective because the color progression corresponds directly to the numerical ordering of satisfaction. It is also more accessible and avoids introducing an artificial visual distinction between values below and above the midpoint.